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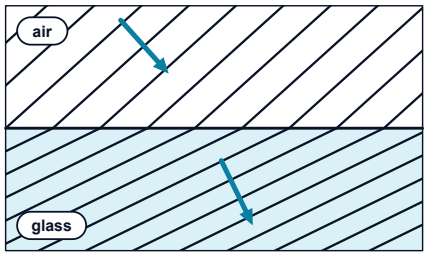
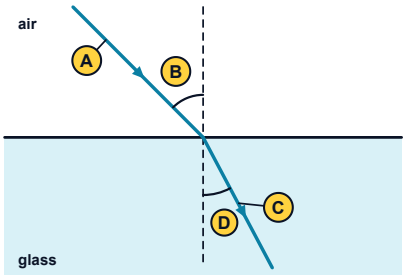
Electromagnetic waves

Waves
Paper 2, Combined and Separate
32 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 16 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Light passes from air into glass. Compare the angle of refraction with the angle of incidence.**	It is smaller (the ray bends towards the normal).	☐
2	**Light passes from glass into air. Compare the angle of refraction with the angle of incidence.**	It is larger (the ray bends away from the normal).	☐
3 ★	**What type of wave are electromagnetic waves?**	Transverse	☐
4	**State what oscillates in electromagnetic waves.**	Electric and magnetic fields	☐
5	**Explain why electromagnetic waves can travel in a vacuum.**	They are oscillations of fields, not particles, so they need no medium.	☐
6 ★	**Name the seven groups of the electromagnetic spectrum, from longest to shortest wavelength.**	Radio waves, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays	☐
7	**Name the electromagnetic waves with the longest wavelength.**	Radio waves	☐
8	**Name the electromagnetic waves with the shortest wavelength.**	Gamma rays	☐
9	**Name the electromagnetic waves that human eyes can detect.**	Visible light	☐
10 ★	**Define refraction.**	The change in direction of a wave at a boundary between two media.	☐
11 ★	**Give two situations in which a wave crossing a boundary does not change direction.**	When it travels along the normal. When its speed is the same in both media.	☐

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	QUESTION	ANSWER	✓
12 ★	**Explain why refraction happens.**	The wave changes speed in the new medium.	☐
13	**Name the process shown in this wavefront diagram.** 	Refraction	☐
14	**State what labels A to D show in this diagram.** 	A: incident ray B: angle of incidence C: refracted ray D: angle of refraction	☐
15 ★	**Describe how radio waves can be produced.**	By oscillations in an electrical circuit.	☐
16 ★	**Describe how radio waves can be detected.**	They are absorbed by an aerial, creating an alternating current of the same frequency.	☐
17 ★	**State two hazards of ultraviolet radiation.**	Premature skin ageing and a higher risk of skin cancer	☐
18 ★	**State the hazards of X-rays and gamma rays.**	They are ionising: they can cause gene mutations and cancer.	☐
19	**State what radio waves are used for.**	Television and radio signals	☐
20 ★	**State what microwaves are used for.**	Satellite communications and cooking food	☐
21 ★	**State what infrared radiation is used for.**	Electrical heaters, cooking food and infrared cameras	☐
22	**State what visible light is used for.**	Fibre optic communications	☐

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	QUESTION	ANSWER	✓
23 ★	**State what ultraviolet radiation is used for.**	Energy-efficient lamps and sun tanning	☐
24 ★	**State what X-rays and gamma rays are used for.**	Medical imaging and treatments	☐
25 ★	**State what all electromagnetic waves have in common in a vacuum.**	They travel at the same speed.	☐
26	**Name the electromagnetic waves with the highest frequency.**	Gamma rays	☐
27	**State what electromagnetic waves transfer.**	Energy, from a source to an absorber.	☐
28	**State the kind of spectrum that electromagnetic waves form.**	A continuous spectrum	☐
29	**State where gamma rays originate.**	From changes in the nucleus of an atom.	☐
30 ★	**Explain why microwaves are suitable for satellite communications.**	They pass through the atmosphere without being absorbed or reflected.	☐
31 ★	**Explain why X-rays are suitable for producing images of bones.**	They pass through soft tissue but are absorbed by bone.	☐
32	**State two things that the hazard from UV, X-rays and gamma rays depends on.**	The type of radiation and the size of the dose	☐

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